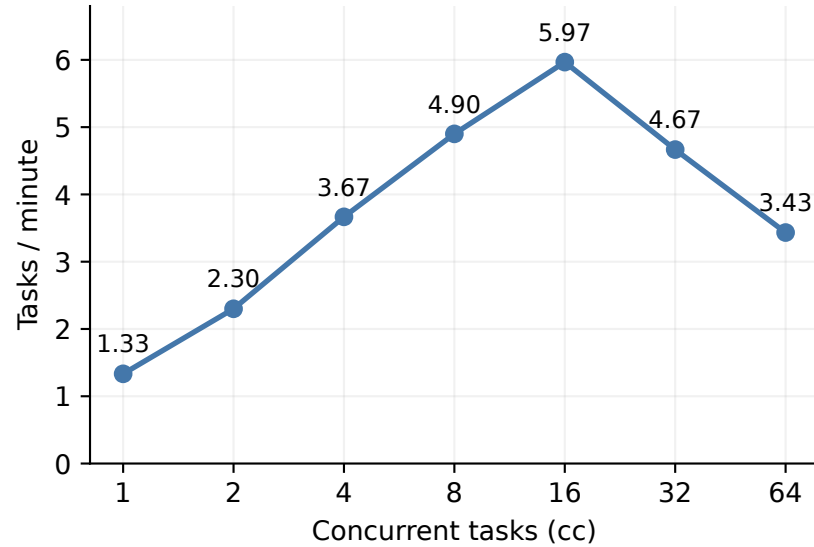
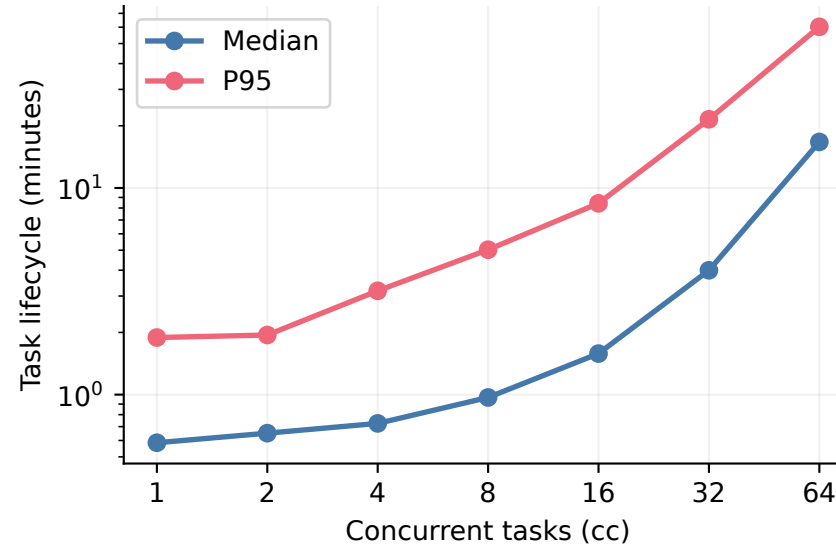


OpenClaw scaling | Qwen3.6-35B-A3B, TP4, 110-trace pool

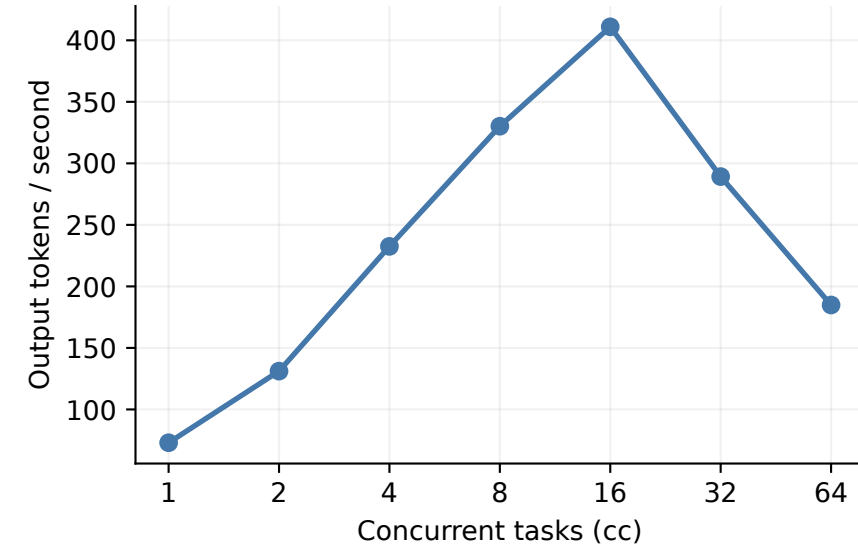
(a) Task throughput



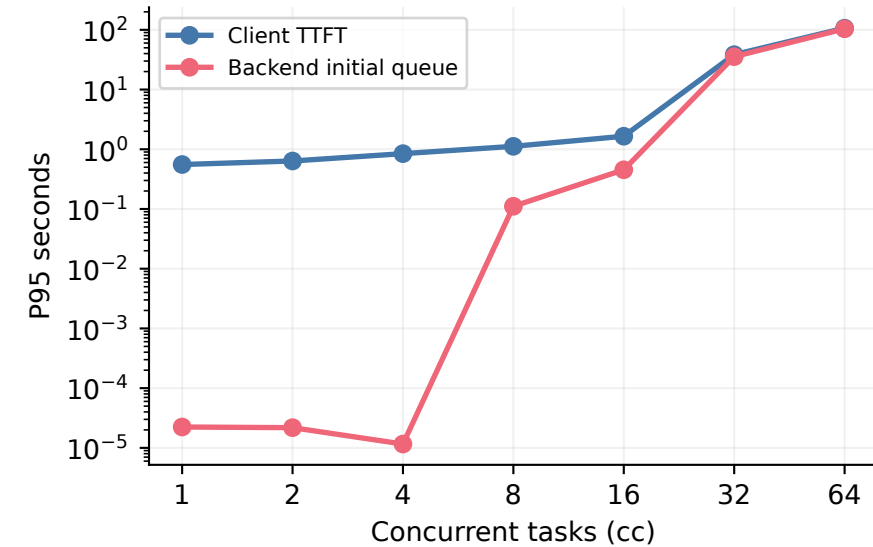
(b) Task latency



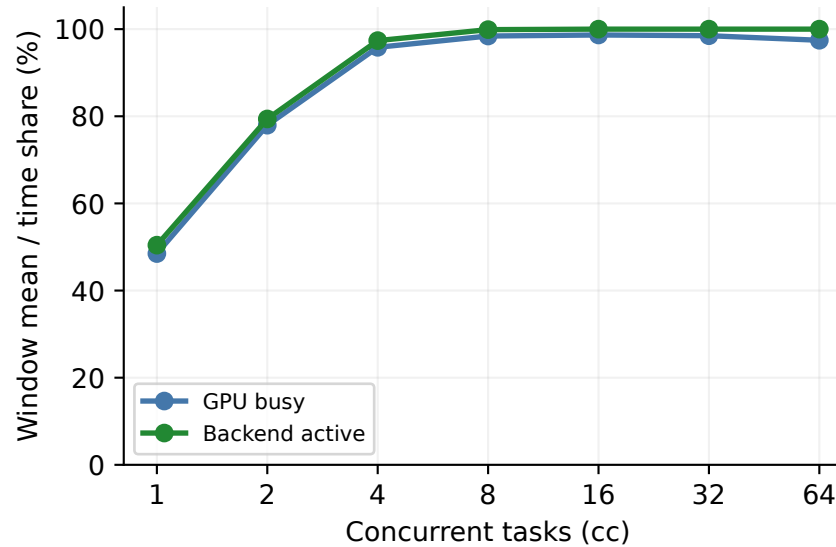
(c) Inference output throughput



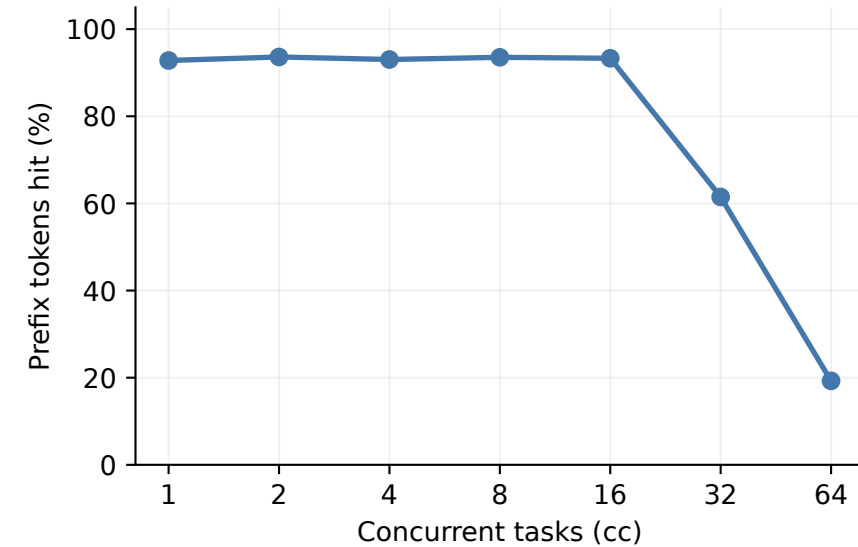
(d) First-token and queue delay



(e) Inference utilization



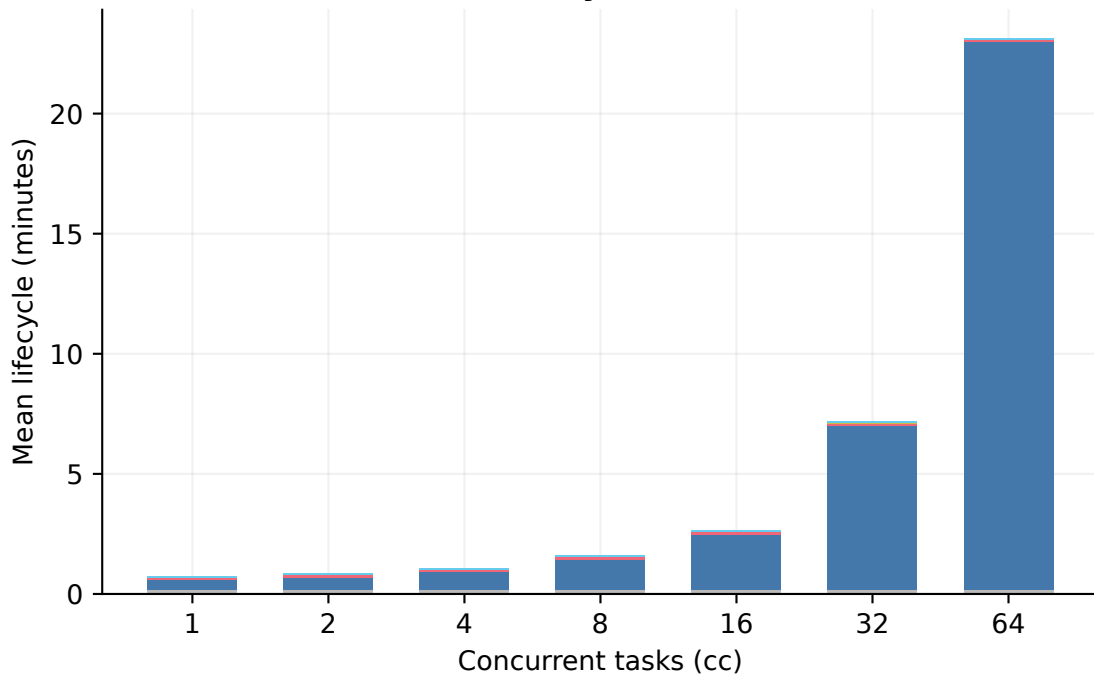
(f) Prefix-cache reuse



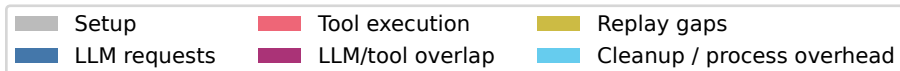
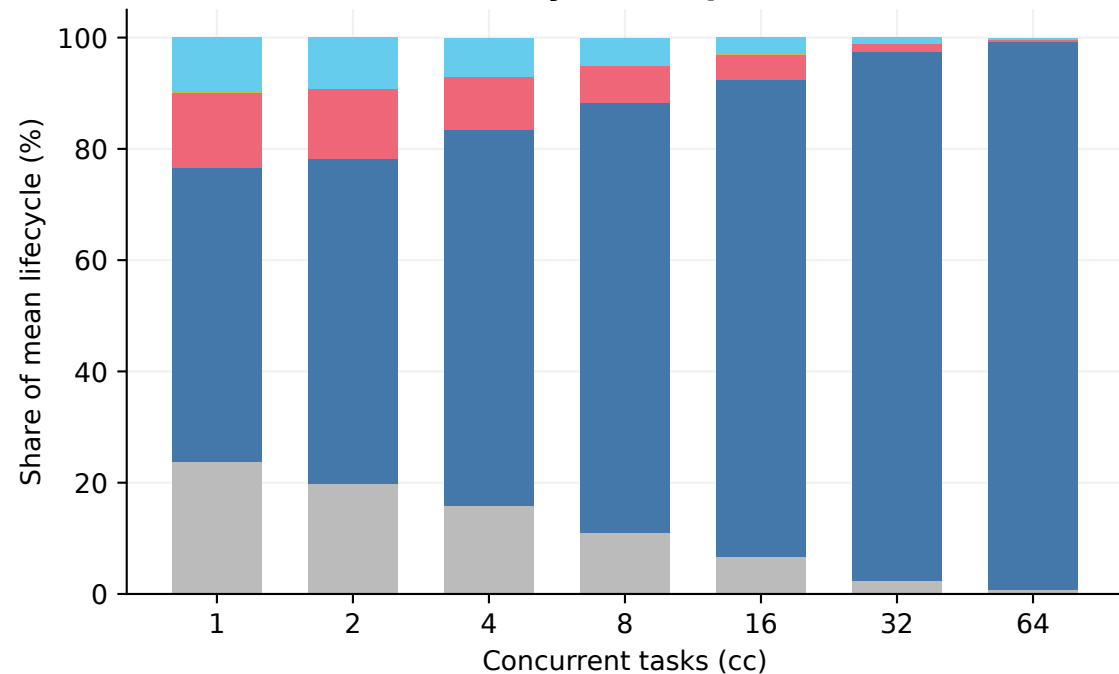
120 s warmup + 1800 s measurement; one repetition per cc. Latencies follow window admissions through natural drain.
web_search uses recorded delay. cc16 is the best observed point in this sweep, not an SLO capacity bound.

Where task wall time is spent | disjoint client intervals

(a) Mean lifecycle breakdown



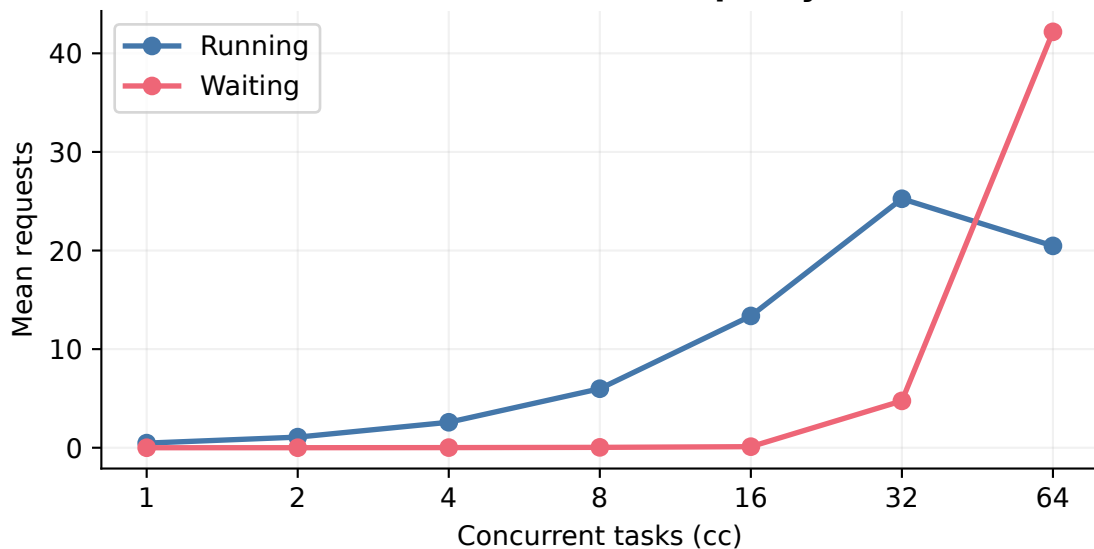
(b) Lifecycle composition



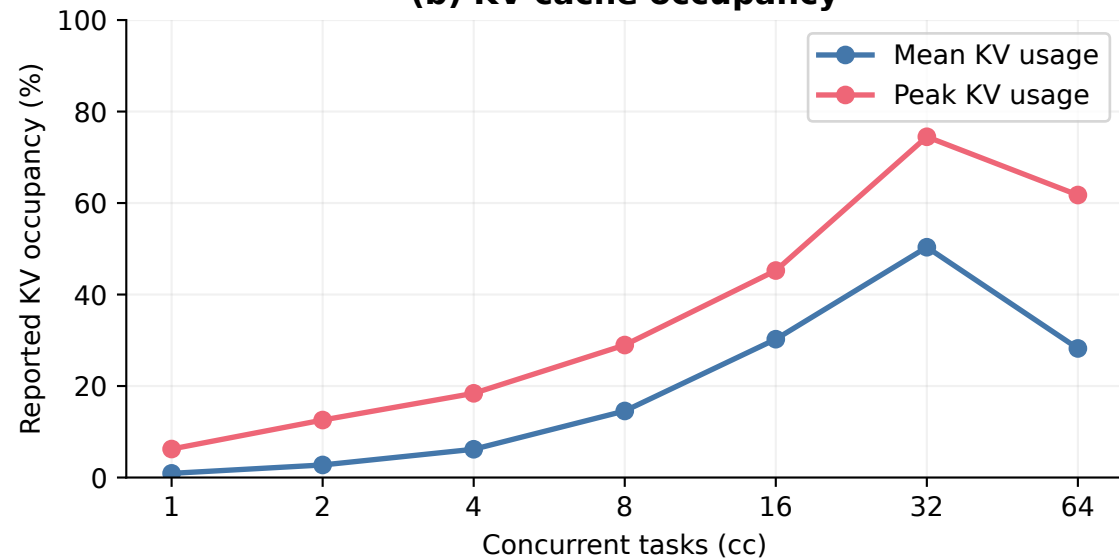
LLM intervals include server waiting and network time. Parallel calls are counted by interval union; overlap is separate.

Backend scheduling, cache and output streaming

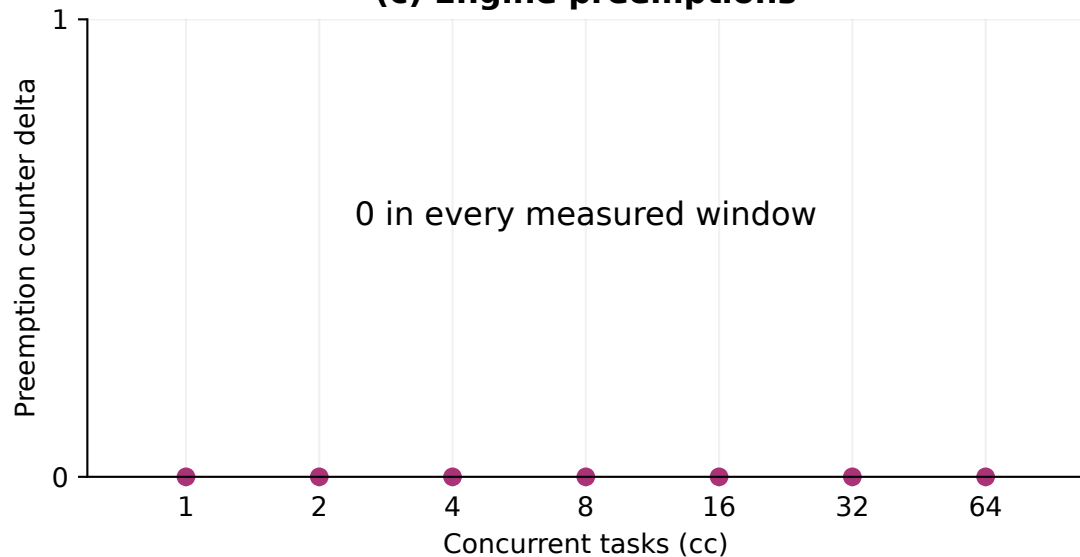
(a) Scheduler occupancy



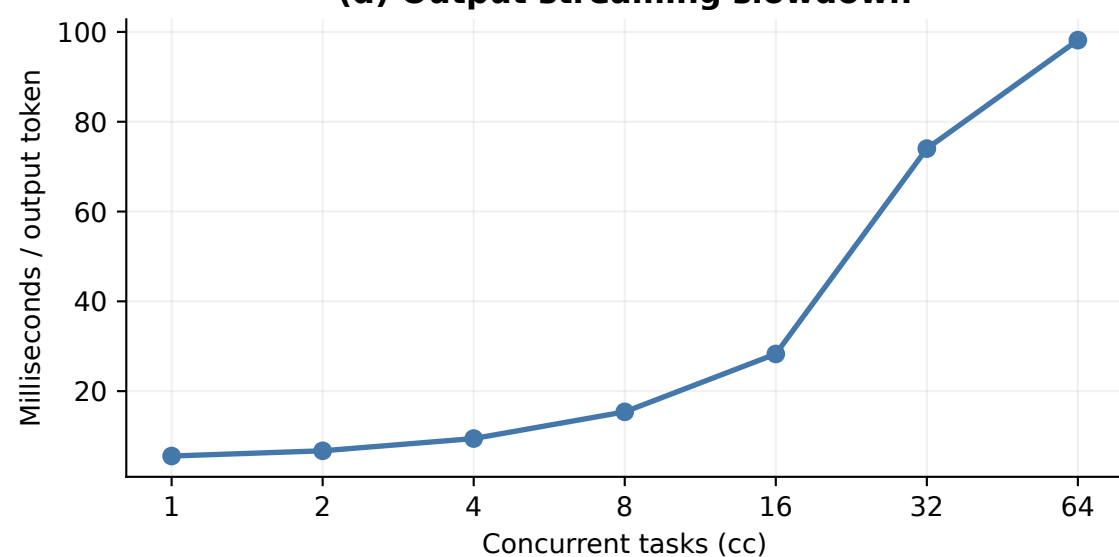
(b) KV-cache occupancy



(c) Engine preemptions



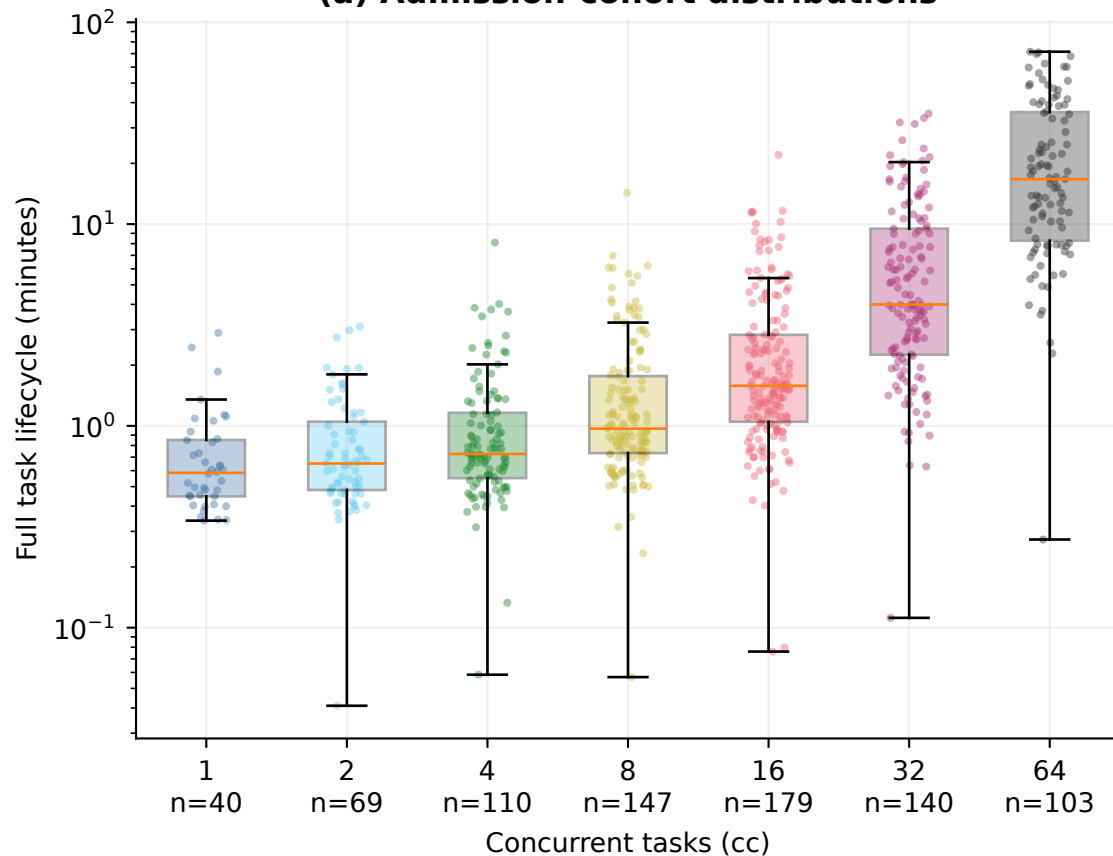
(d) Output streaming slowdown



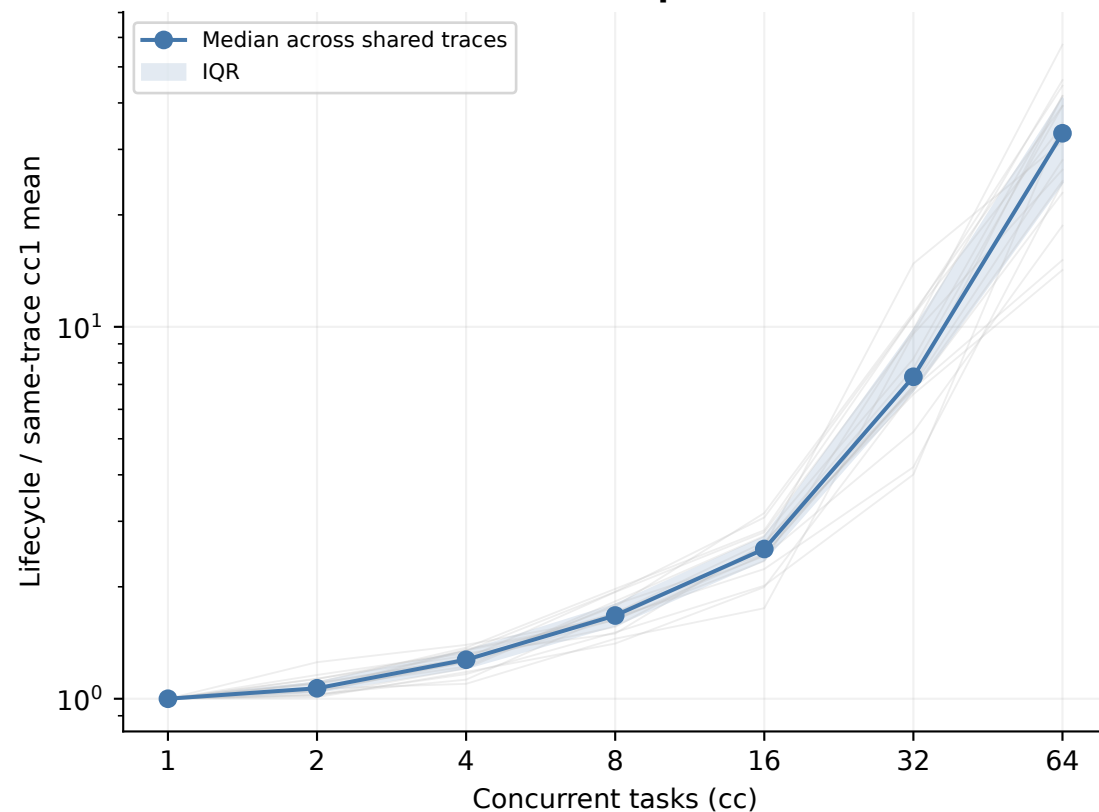
KV occupancy is the reported vLLM gauge, not total allocated GPU memory. Counters span the in-window samples (~1799 s).
TPOT is estimated from client streaming chunks; it is not an exact per-token engine measurement.

Latency distributions and workload-mix sensitivity

(a) Admission-cohort distributions



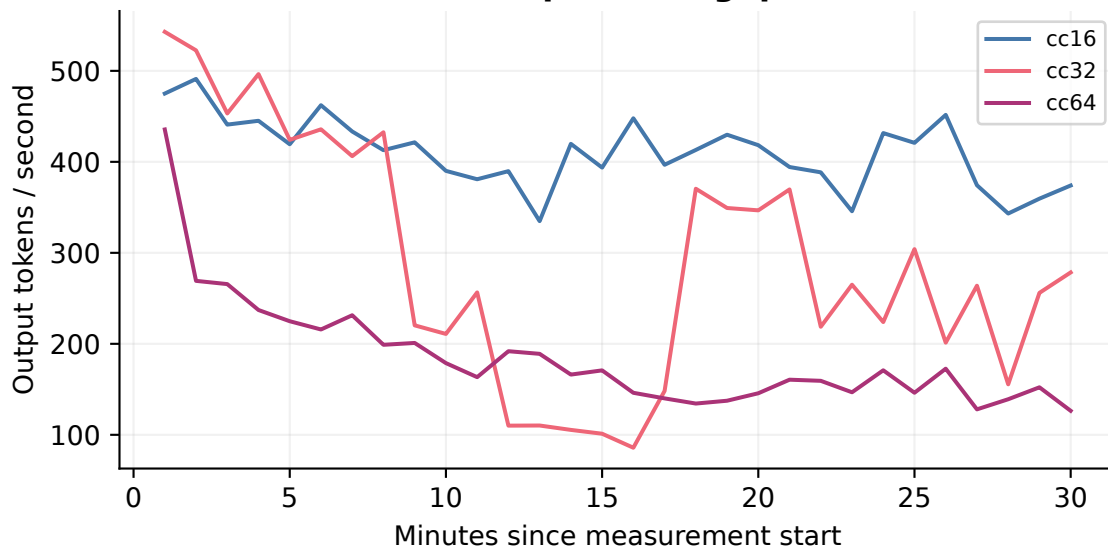
(b) Same-trace comparison (n=18)



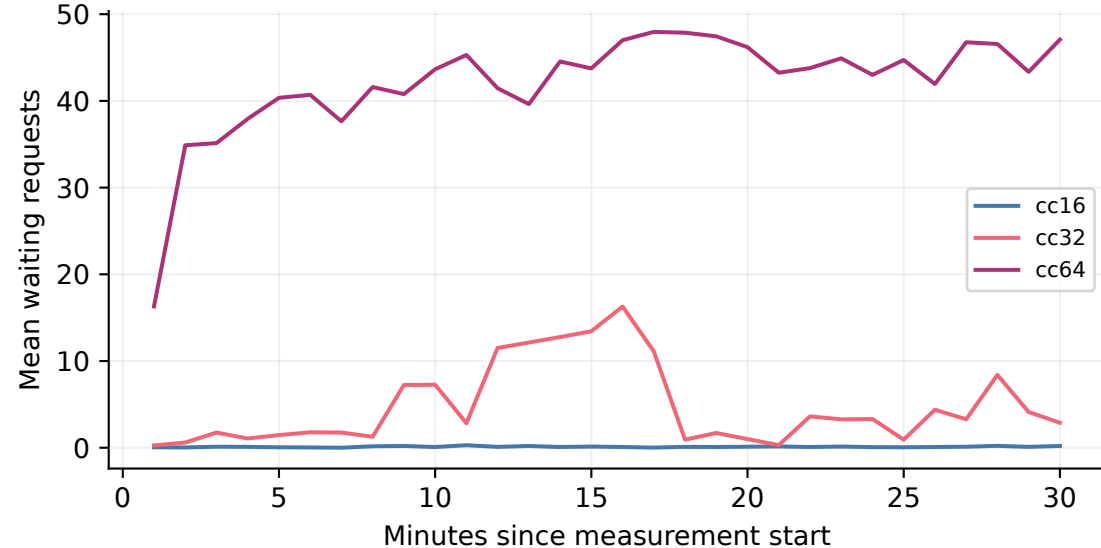
Boxes: median / IQR, whiskers 1.5 IQR; every observed task is shown. No censored task latency is dropped.
Paired comparison averages repeats per trace, then weights shared traces equally; it does not control cache history or tool outcomes.

Within-window trends |cc16, cc32 and cc64

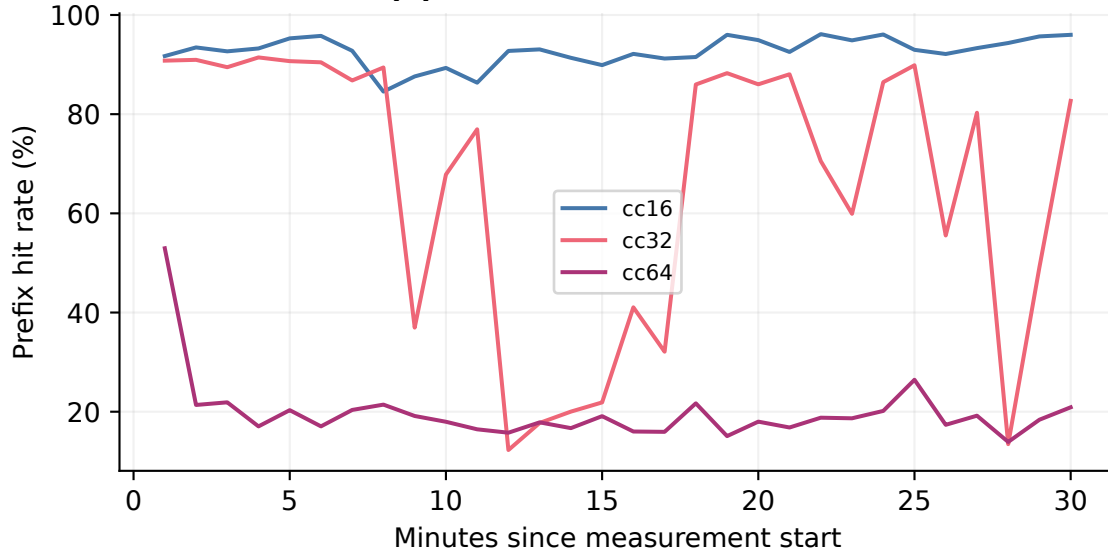
(a) Output throughput



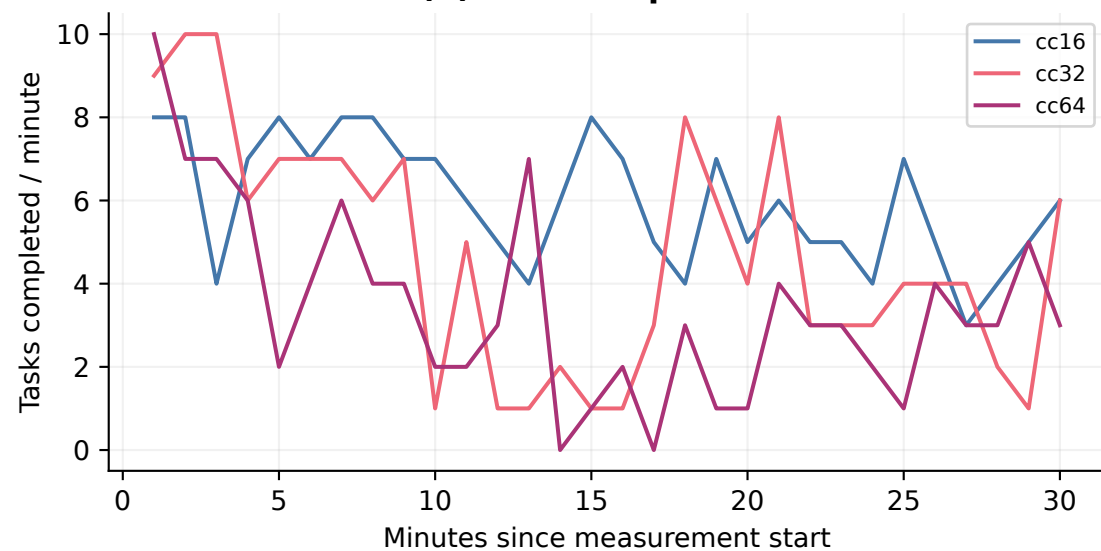
(b) Queue accumulation



(c) Interval cache hit rate



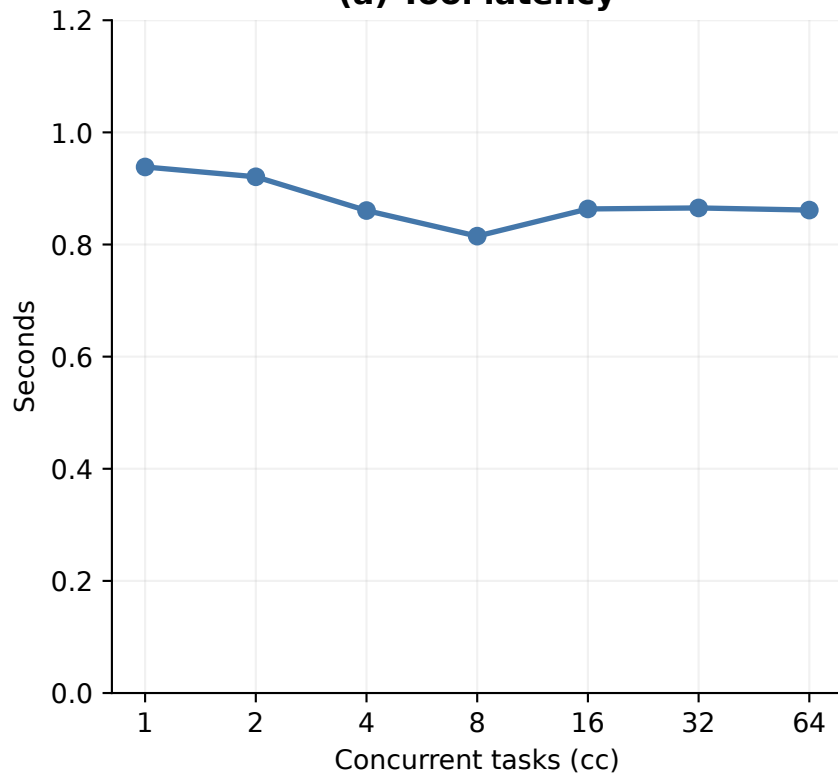
(d) Task completions



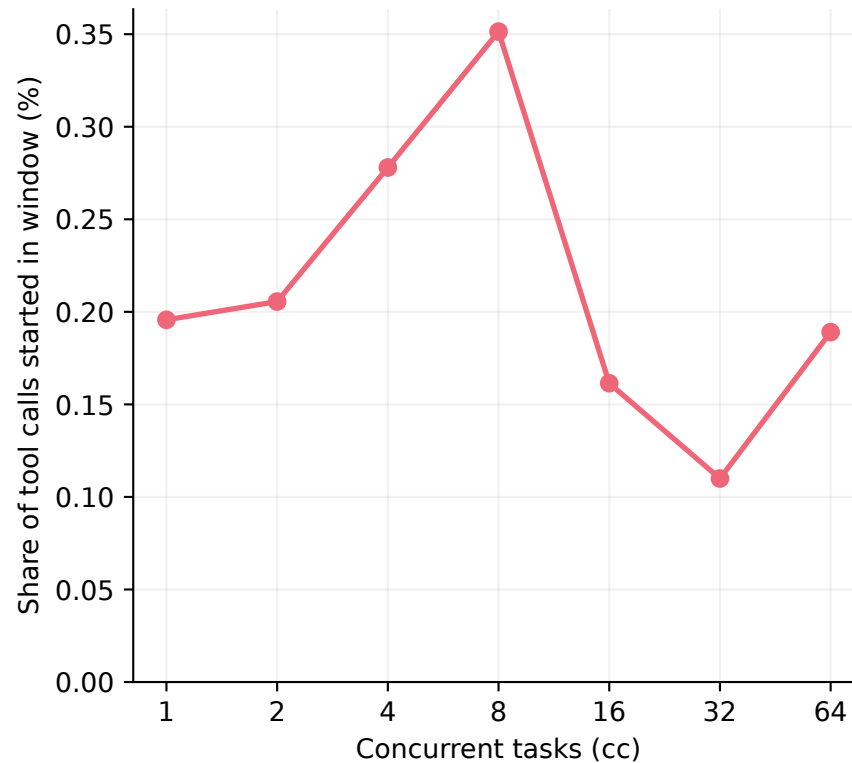
60-second bins; token/cache rates use sampled counter deltas within each bin (~59 s). Task counts use full 60 s bins.
These curves assess transient behavior; a 30-minute measurement does not by itself establish steady state.

Tool behavior and sample coverage

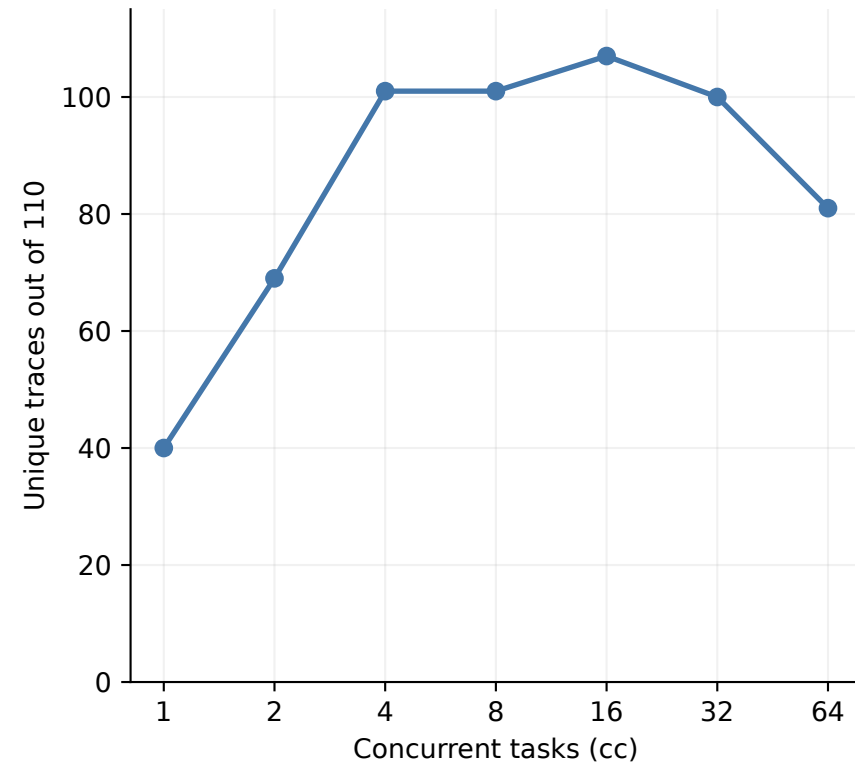
(a) Tool latency



(b) Newly failing tool calls



(c) Admission-cohort coverage



Native tool failures are retained; engine validation does not certify identical tool outcomes.
All points use the same trace pool and seed, but complete different task cohorts within a fixed window.